

SWaP-C Down-Selection Report

One-Way Attack (OWA) Interdiction (C-sUAS) | CENTCOM | Posture: Attributable First (default) | Generated 26 Jul 2026 17:56Z

BLUF: General Cherry Bullet is the primary recommendation for One-Way Attack (OWA) Interdiction (C-sUAS) in CENTCOM, classified T0 Consumable / S1 Man-Packable at \$3,000 per unit. Cost per defeat is \$7,500 for a 5.3:1 exchange against a \$40,000 value denied. Against the Raytheon Coyote Block 2 baseline it is 21.4x cheaper per defeat (\$7,500 versus \$160,256). Alternate effectors: Merops / AS3 Surveyor, TYTAN METIS. Pair with Acoustic Sensor Mesh Node (Sky Fortress pattern) at \$600 per node for detection and cueing; attributable effectors need an external sensor picture to be employable.

Input	Value
Mission thread	One-Way Attack (OWA) Interdiction (C-sUAS)
AOR	U.S. Central Command (CENTCOM)
Cost per unit threshold	\$600,000
Flight time requirement	5 min
Distance requirement	8 km
Scoring posture	Attributable First (default) — Maximize favorable cost exchange and magazine depth. Penalize exquisite systems.
Cost tier ceiling	T4
Reference threat cost	\$40,000
Exchange ratio target	8:1
Assumed salvo size	20 threats
Passing candidates by tier	T0: 1, T1: 3, T2: 2

Ranked Effectors (Attributable Set)

#	System	Tier	Score	Unit Cost	Per Defeat	Exch	SWaP	Prod/mo	Evidence
1	General Cherry Bullet General Cherry	T0 S1	0.85	\$3,000	\$7,500	5.3:1	8 lb 200 W	20,000	combat proven
2	Merops / AS3 Surveyor Perennial Autonomy	T1 S1	0.72	\$15,000	\$25,000	1.6:1	15 lb 300 W	4,000	combat proven
3	TYTAN METIS TYTAN Technologies	T1 S1	0.69	\$18,000	\$31,034	1.3:1	13 lb 300 W	3,000	combat proven
4	APKWS II / VAMPIRE BAE Systems / L3Harris	T2 S2	0.60	\$30,000	\$42,857	0.9:1	32 lb 1,500 W	2,000	combat proven
5	Frankenburg Mark I Frankenburg Technologies	T1 S1	0.58	\$25,000	\$35,714	1.1:1	18 lb 200 W	3,000	demonstrated
6	Alpine Eagle Sentinel Alpine Eagle	T2 S2	0.49	\$35,000	\$70,000	0.6:1	40 lb 1,200 W	300	demonstrated

1. General Cherry Bullet — T0 Consumable / S1 Man-Packable \$7,500 per defeat, 5.3:1 exchange Favorable at 5.3:1 against a \$40,000 threat.

2. **Merops / AS3 Surveyor** — T1 Attributable / S1 Man-Packable \$25,000 per defeat, 1.6:1 exchange Marginally favorable at 1.6:1. Thin margin against cheaper threats.
3. **TYTAN METIS** — T1 Attributable / S1 Man-Packable \$31,034 per defeat, 1.3:1 exchange Marginally favorable at 1.3:1. Thin margin against cheaper threats.

Detection and Cueing Layer (Enablers)

These are not substitutes for an effector and are ranked separately. Attributable interceptors are only employable with an external sensor picture, so the architecture needs at least one of these.

System	Tier	Unit Cost	Range	SWaP	Note
Acoustic Sensor Mesh Node (Sky Fortress pattern) Multiple / reference architecture	T0/S1	\$600	3.0 km	4 lb / 15 W	Detection only, no defeat capability. The cheapest way to buy wide-area early warning and cue expensive effectors. Passive and emission-free, so it survives EW and does not violate spectrum constraints. Thousands of nodes cost less than one exquisite radar.

Program-of-Record Baseline Comparison

These systems are excluded from the ranked set by design. They are shown so the cost-exchange delta of the attributable recommendation is explicit, and so any capability genuinely unique to them is visible.

System	Tier	Unit Cost	Per Defeat	Exch	Delta vs Top	Note
Raytheon Coyote Block 2	T3	\$125,000	\$160,256	0.25:1	21.4x costlier	POR baseline comparator. Highest Pk in the kinetic set with a mature sustainment base, but unit cost is roughly 8x Merops for a comparable engagement. Long-term production framework agreement in place with RTX.
Anduril Roadrunner-M	T3	\$500,000	\$147,059	0.27:1	19.6x costlier	POR baseline comparator. Recoverability amortizes the high unit cost across sorties, so effective cost per engagement beats the sticker price. Still an order of magnitude above attributable options.
Epirus Leonidas	T4	\$12,000,000	\$1,412	28.33:1	0.2x costlier	POR baseline comparator that inverts the usual math: worst unit cost in the catalog, best cost per engagement once amortized across a deep magazine. Short range confines it to terminal defense. The exchange model captures this, whereas unit-cost filtering alone would wrongly discard it.

Magazine Depth and Replenishment

Sized against a 20-threat salvo at 400 estimated engagements per month in CENTCOM.

System	Rounds/Defeat	Magazine Rounds	Magazine Cost	Replenish (days)
General Cherry Bullet	2.50	50	\$150,000	0.1
Merops / AS3 Surveyor	1.67	34	\$500,000	0.2
TYTAN METIS	1.72	35	\$620,690	0.3
APKWS II / VAMPIRE	1.43	29	\$857,143	0.4
Frankenburg Mark I	1.43	29	\$714,286	0.3
Alpine Eagle Sentinel	2.00	40	\$1,400,000	4.0

Sensitivity Analysis

Deterministic 3x3 grid: unit cost x0.5/x1.0/x1.5, Pk -0.15/0/+0.15. Rank ranges show each candidate's best and worst position across all nine scenarios.

general_cherry_bullet holds rank 1 in 9 of 9 perturbation scenarios. The recommendation is robust to the stated data uncertainty.

System	Rank Range	Verdict
General Cherry Bullet	1-1	Stable
Merops / AS3 Surveyor	2-2	Stable

TYTAN METIS	3-3	Stable
APKWS II / VAMPIRE	4-4	Stable
Frankenburg Mark I	5-5	Stable
Alpine Eagle Sentinel	6-6	Stable

Layered Architecture Proposal

3-layer architecture assuming one engagement per layer. Leakage 10.1%: of a 20-threat salvo, expect about 2.0 leakers reaching the terminal point, before reattack. Deepen magazines or add a layer to drive this down.

Layer	Band (km)	System	Effective Pk	Magazine Cost
Outer	10+	Merops / AS3 Surveyor	0.60	\$500,000
Mid	3-10	General Cherry Bullet	0.40	\$150,000
Terminal	0-3	TYTAN METIS	0.58	\$620,690
Sensor	-	Acoustic Sensor Mesh Node (Sky Fortress pattern)	-	\$600

Total architecture magazine cost: **\$1,271,290. Leakage: 10.1%.**

Watch Items

General Cherry Bullet

- Cost is an order-of-magnitude placeholder (2026-05), not a quote. Source: Vendor claims capacity scalable to 100,000 drones per month. Unit cost inferred from the FPV interceptor class, not published.

TYTAN METIS

- Cost is an order-of-magnitude placeholder (2026-06), not a quote. Source: Performance from vendor claims: over 250 km/h, over 15 km engagement range. Unit cost inferred, not published.

- Startup vendor. Confirm production capacity, financial runway, and second-source options before committing to a magazine.

APKWS II / VAMPIRE

- Unfavorable cost exchange at 0.93:1 against a \$40,000 threat.

Frankenburg Mark I

- Cost is an order-of-magnitude placeholder (2026-02), not a quote. Source: Live-fire tests Aug and Dec 2025. Cost not published; positioned as a low-cost interceptor missile. Production goal over 100 missiles per day per site.

- Startup vendor. Confirm production capacity, financial runway, and second-source options before committing to a magazine.

Alpine Eagle Sentinel

- Cost is an order-of-magnitude placeholder (2026-05), not a quote. Source: Interceptor unit cost not published. Ukraine trials and Project Vanaheim participation reported.

- Startup vendor. Confirm production capacity, financial runway, and second-source options before committing to a magazine.

- Unfavorable cost exchange at 0.57:1 against a \$40,000 threat.

Threat Reference

Group 2 - Small: ScanEagle-class, Orlan-10, Lancet-class loitering munition, heavy-lift quadcopter. Median threat cost \$60,000; affordable defeat ceiling \$20,000. Fixed-wing ISR and heavier attack payloads. Faster and higher than Group 1, stressing detection timelines for point defense. Threat cost is high enough that T1 attritable interceptors return a strongly favorable exchange.

Group 3 - Medium: Shahed-136 OWA, Mohajer-6, TB2-class. Median threat cost \$40,000; affordable defeat ceiling \$15,000. OWA cruise-missile-like profiles and armed ISR. Note the cost anomaly: Shahed-class OWA sits at the low end of this group's cost band at \$30-50K despite Group 3 performance, which is precisely why million-dollar interceptors lose. Handheld EW is ineffective; kinetic or HPM required.

Retrieved Knowledge Base Context

One-Way Attack (OWA) Interdiction (C-sUAS) [mission] (relevance 364.9): One-Way Attack (OWA) Interdiction (C-sUAS). Targets groups (2, 3). Salvo 20, exchange target 8.0:1, attritability critical. Defeat of Shahed-class OWA UAS on cruise-missile-like profiles. Long-range detection, high-speed intercept, and magazine depth against salvo attacks. This is the thread where cost exchange is most decisive: the threat costs \$30-50K and arrives in tens. Requirements high_speed_intercept magazine_depth long_range_detect low_cost_per_intercept gps_denied_ops high_production_rate.

U.S. Central Command [threat] (relevance 341.8): U.S. Central Command Middle East, Central Asia, Egypt. Highest OPTEMPO for one-way attack UAS and Group 1-3 ISR threats. Iranian-aligned proxy employment of Shahed-class OWA and quadcopter FPV against fixed sites and maritime assets. GPS jamming and spoofing routine. Gulf reporting suggests a larger share of drones penetrate defenses than ballistic missiles.

Group 3 - Medium [threat] (relevance 163.5): Group 3 - Medium. Max weight 1320.0 lb, endurance 240.0-900.0 min, range 80.0-1500.0 km. Median threat cost \$40,000, affordable defeat ceiling \$15,000. Threats Shahed-136 OWA, Mohajer-6, TB2-class. OWA cruise-missile-like profiles and armed ISR. Note the cost anomaly: Shahed-class OWA sits at the low end of this group's cost band at \$30-50K despite Group 3 performance, which is precisely why million-dollar interceptors lose. Handheld EW is ineffective; kinetic or HPM required.

Group 2 - Small [threat] (relevance 143.5): Group 2 - Small. Max weight 55.0 lb, endurance 60.0-240.0 min, range 15.0-80.0 km. Median threat cost \$60,000, affordable defeat ceiling \$20,000. Threats ScanEagle-class, Orlan-10, Lancet-class loitering munition, heavy-lift quadcopter. Fixed-wing ISR and heavier attack payloads. Faster and higher than Group 1, stressing detection timelines for point defense. Threat cost is high enough that T1 attritable interceptors return a strongly favorable exchange.

Fixed-Site Base Defense (C-sUAS) [mission] (relevance 133.6): Fixed-Site Base Defense (C-sUAS). Targets groups (1, 2, 3). Salvo 12, exchange target 5.0:1, attritability critical. Layered detect-track-identify-defeat of Group 1-3 threats against an installation. Persistent 360-degree coverage with integrated C2 (FAAD C2 class). Saturation OWA and FPV raids make low cost per intercept the design driver, not single-shot performance. Requirements persistent_coverage layered_defeat c2_integration low_cost_per_intercept saturation_resilience magazine_depth.

T1 Attritable [taxonomy] (relevance 131.5): T1 Attritable up to \$25,000 per unit. Expendable against any Group 1-3 threat. Favorable exchange against Shahed-class OWA. Buy in hundreds to thousands. Field-replaceable, no component-level repair.

U.S. European Command [threat] (relevance 113.9): U.S. European Command Europe, Russia, Greenland, Iceland. Peer EW threat informed by Ukraine lessons. FPV, Lancet-class loitering munitions, Orlan-class ISR, and Shahed-class OWA at scale. Russia reportedly launched over 54,000 Shahed-type drones by early 2026. Dense friendly airspace deconfliction burden.

Group 1 - Micro/Mini [threat] (relevance 99.7): Group 1 - Micro/Mini. Max weight 20.0 lb, endurance 15.0-90.0 min, range 5.0-15.0 km. Median threat cost \$3,000, affordable defeat ceiling \$3,000. Threats DJI Mavic/Matrice, FPV racing quads, fiber-optic FPV, Autel EVO. Highest-density threat class worldwide and the hardest exchange problem: a \$500 fiber-optic FPV cannot be economically met with any kinetic interceptor. Reusable EW, smart optics, and acoustic cueing are the only effectors that win the math. Fiber and autonomous variants defeat RF-only defeat systems entirely.

UNCLASSIFIED open-source estimates. Unit costs in the attritable segment move fast and many are vendor claims or single-source press figures. Every entry carries cost_confidence and as_of fields. Treat order_of_magnitude entries as planning placeholders only and refresh against vendor quotes and program-office data before any acquisition decision.